Proof. By the Maschke theorem, $k[G]$ is semisimple, so by Theorem 3.6.2, the characters are linearly independent and are a basis of $(A/[A,A])^*$, where $A = k[G]$. It suffices to note that, as vector spaces over k,

$$\begin{aligned}(A/[A,A])^* &\cong \{\varphi \in \mathrm{Hom}_k(k[G],k) \mid gh - hg \in \ker\varphi \ \forall g,h \in G\}\\ &\cong \{f \in F(G,k) \mid f(gh) = f(hg) \ \forall g,h \in G\},\end{aligned}$$

which is precisely $F_c(G,k)$. □

Corollary 4.2.2. *The number of isomorphism classes of irreducible representations of G equals the number of conjugacy classes of G (if $|G| \neq 0$ in k).*

Exercise 4.2.3. Show that if $|G| = 0$ in k, then the number of isomorphism classes of irreducible representations of G over k is strictly less than the number of conjugacy classes in G.

Hint: Let $P = \sum_{g\in G} g \in k[G]$. Then $P^2 = 0$. So P has zero trace in every finite dimensional representation of G over k.

Corollary 4.2.4. *Any finite dimensional representation of G is determined by its character if k has characteristic 0; namely, $\chi_V = \chi_W$ implies $V \cong W$.*

4.3. Examples

The following are examples of representations of finite groups over $\mathbb{C}$.

(1) Finite abelian groups $G = \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}$. Let $G^\vee$ be the set of irreducible representations of G. Every element of G forms a conjugacy class, so $|G^\vee| = |G|$. Recall that all irreducible representations over $\mathbb{C}$ (and algebraically closed fields in general) of commutative algebras and groups are 1-dimensional. Thus, $G^\vee$ is an abelian group: if $\rho_1, \rho_2 : G \to \mathbb{C}^\times$ are irreducible representations, then so are the representations $\rho_1(g)\rho_2(g)$ and $\rho_1(g)^{-1}$. The group $G^\vee$ is called the **dual group** or **character group** of G.

For given $n \geq 1$, define $\rho : \mathbb{Z}_n \to \mathbb{C}^\times$ by $\rho(m) = e^{2\pi i m/n}$. Then $\mathbb{Z}_n^\vee = \{\rho^k : k = 0, \ldots, n-1\}$, so $\mathbb{Z}_n^\vee \cong \mathbb{Z}_n$. In general,

$$(G_1 \times G_2 \times \cdots \times G_n)^\vee = G_1^\vee \times G_2^\vee \times \cdots \times G_n^\vee,$$